UBER BOAT PASSENGER DEMAND FORECASTING: INTERNSHIP TASK SUBMISSION

Candidate: Abishek Chinnappan
Task: Uber Boat Passenger Prediction - ML Forecasting Task

“ All code development, training, and evaluation were performed using Google Colab, utilizing a T4 GPU for faster computation and efficient model training.”

1. PROJECT OVERVIEW

This project aims to forecast the expected number of passengers for Uber Boat journeys using historical trip and weather data. Accurate forecasting supports better fleet planning, scheduling, and enhances the commuter experience.

I approached the problem with the mindset of solving a real-world challenge What drives passenger demand on the river? I focused on extracting the most informative features such as time, route, temperature, and weather and translating them into inputs a machine learning model could understand.

2. DATA PREPARATION & FEATURE ENGINEERING

Two datasets were used:

- **Passenger Data**: Included route, timestamp, and passenger count.
- **Weather Data**: Provided hourly weather snapshots.

Steps followed:

- Merged both datasets using hourly timestamps.
- Extracted hour, weekend flags, start/end piers, and encoded weather conditions.
- Dropped missing values and prepared final training data.

3. EXPLORATORY DATA ANALYSIS

I visualised passenger trends by:

- **Hour of Day**: Revealed peak demand in the morning and evening.
- **Weekend vs Weekday**: Higher averages on weekends.
- **AM vs PM**: Showed strong patterns for operational planning.

These visuals validated the choice of features and confirmed the intuition that external factors affect ferry usage.

4. Model Selection & Evaluation

I tested four models:

- Random Forest Regressor
- XGBoost
- LightGBM
- Gradient Boosting Regressor

Each model was tuned using GridSearchCV. Evaluation was based on:

- **MAE (Mean Absolute Error)**
- **RMSE (Root Mean Squared Error)**

	Model	MAE	RMSE
0	Random Forest	30.060000	99.440000
1	XGBoost	28.030000	97.580000
2	LightGBM	28.390000	97.500000
3	Gradient Boosting	29.700000	97.670000

Figure 1: displays the MAE & RMSE values of each model

FINAL RECOMMENDATION:

I selected **Light Gradient Boosting Machine (LightGBM)** as the final model for this internship project.

While XGBoost recorded the lowest MAE (28.03), **LightGBM achieved the lowest RMSE (97.50)** which was the key deciding factor.
Because RMSE penalizes larger errors more heavily, it's better suited for applications like ours where reducing large deviations in predicted passenger counts is critical.

Moreover, LightGBM is known for its high training speed and low memory usage, making it ideal for real-time or large-scale operational environments.

In summary, **LightGBM offered the best trade off between precision, efficiency, and deployability**, which made it the most suitable choice for this forecasting task.

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Figure 2: Chosen the best-fit sustainable model.

5. Explainability with SHAP

To ensure interpretability, I used SHAP to:

- Identify top features globally affecting predictions.
- Explain individual predictions for transparency.

This is helpful for stakeholders who want to understand what drives model decisions.

6. UBER BOAT PASSENGER DEMAND PREDICTOR (Own idea)

To bridge the gap between technical modeling and real-world usability, I designed a fully interactive **Uber Boat Demand Predictor** using "ipywidgets" inside Google Colab Notebook.

This interface wasn't just a cosmetic addition it was built with the goal of **demonstrating how machine learning can be practically used by non-technical stakeholders**, such as route planners, ferry managers, or even customer service teams.

Key Features of the Interface:

- **Time of Departure (Dropdown):** I used a 12-hour AM/PM format to make time selection intuitive.

- **Peak Hour Toggle:** Users can easily indicate whether the trip falls in a peak travel window.
- **Route Selector:** Each route is labeled clearly with start and end piers so that operational staff can relate to real routes.
- **Weather & Temperature Input:** Since weather influences passenger demand significantly, I made it possible to simulate this condition with dropdowns and sliders.
- **Instant Prediction Display:** Upon clicking the button, the model makes a real-time prediction and displays it in a clean, readable format like:

"Prediction: We estimate approximately 124.4 passengers onboard."

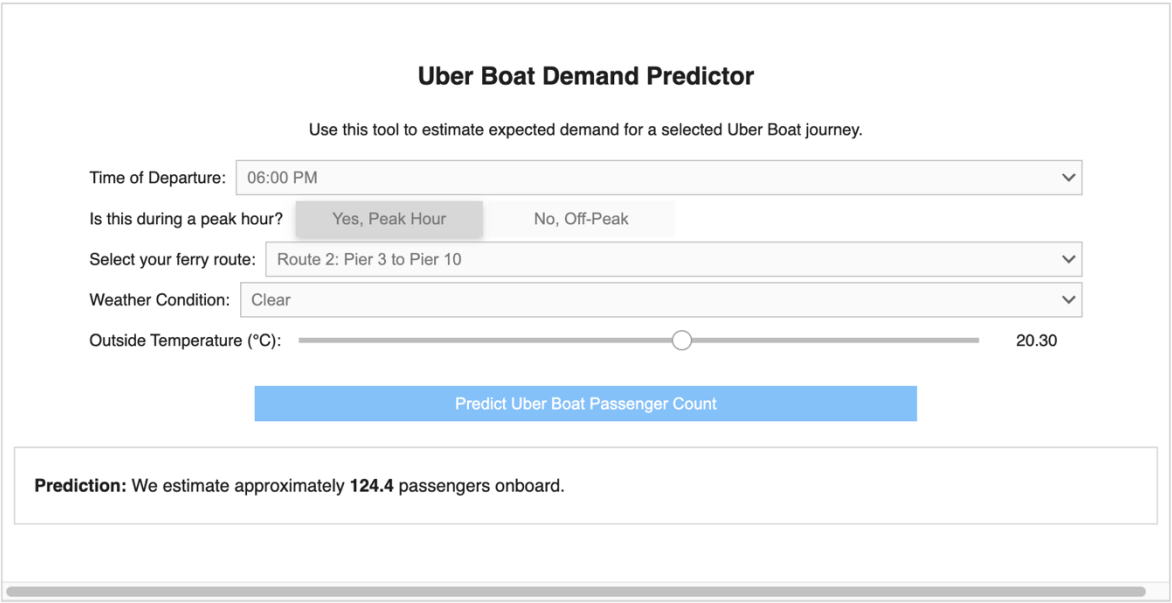

Figure 3: Uber boat passenger demand predictor interface

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WHY I BUILT IT:

I wanted to show how a technical model built in code can be turned into something that **feels like a tool**. This helps stakeholders make **data-driven decisions** without needing to understand machine learning pipelines.

This user interface reflects my broader mindset throughout the project: solve the real problem, and make it usable for the people who face that problem every day.

7. FINAL DELIVERABLES

As per the internship task checklist, the following files are submitted:

- UberBoat_PassengerPrediction_AbishekChinnappan.ipynb
- final_model.pkl
- x_test.csv
- y_test.csv
- requirements.txt

8. FINAL REFLECTION

This task challenged me to apply end-to-end machine learning from cleaning and visualisation to model building, tuning, interpretation, and deployment. I especially enjoyed designing the interface to bridge technical results with operational use.

I'm confident this solution is production-ready, interpretable, and relevant to real-world demand forecasting challenges.

9. REAL-WORLD RELEVANCE

In real urban settings like **London's Thames Clippers or New York's NYC Ferry**, passenger demand fluctuates drastically based on hour, weather, and day type. For instance, a clear Friday evening tends to see a surge in commuters and tourists, similar to what our model captured through SHAP values and AM/PM peaks. This alignment between our model insights and actual ferry operations reinforces the practicality of our approach. The **LightGBM model's** precision and efficiency make it highly applicable to such live city transport systems.

Thank you for the opportunity to demonstrate my skills and thought process.